INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA
Term – I , Second Semester Examination, February- 2025
Subject Code: BCACC201
Subject Name: Data Structures and Algorithms

Full Marks: 30 SET 3 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. Define an array. How is an array stored in memory? 2

2. What are the different types of algorithm complexities? 2

3. What does 'front' signifies in queue. 2

4. What is dynamic memory allocation? 2

5. What is the importance of postfix notation in stack operations? 2

6. What are the advantages of a circular queue over a linear queue? 2

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. How do you test for an empty and full Queue? 5

8. Explain the usage of stack in recursive algorithm implementation? 5

9. Write down algorithms for the insertion and deletion operations performed on circular queue 5

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10. What are the drawbacks of queues? Write an algorithm to insert and delete a data in a circular queue 10

11. What are the postfix and prefix forms of the expression? 10
A+B*(C-D)/(P-R)

--End--